INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS553)

LAB 2

Assignment 2: Largest, Smallest and Count

Assignment aim: Find extrema and count values that satisfy a condition.

Experiment No. 2.1 — Find largest array value

1. Experiment Title: Find largest array value

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A traversal can track the current minimum/maximum or count elements meeting a condition.

5. Algorithm / Procedure:
Assume the first value is largest; compare every remaining value; update largest when a greater value is found.
Then compile and execute the Java program.

6. Program:

```
public class Task_01_Find_largest_array_value {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int largest = marks[0];
        for (int i = 1; i < marks.length; i++) {
            if (marks[i] > largest) largest = marks[i];
        }
        System.out.println("Largest: " + largest);
    }
}
```

7. Input: Fixed input values are declared in the program: marks = {45, 67, 80, 52, 90}.

8. Output:
Largest: 90

Time / Space Complexity: O(n) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- Why start with marks[0]?
- What happens if the array is empty?
- What is the time complexity?

Experiment No. 2.2 — Find smallest array value

1. Experiment Title: Find smallest array value

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A traversal can track the current minimum/maximum or count elements meeting a condition.

5. Algorithm / Procedure:
Assume the first value is smallest; compare the rest; update smallest whenever a lower value appears.
Then compile and execute the Java program.

6. Program:

```
public class Task_02_Find_smallest_array_value{
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int smallest = marks[0];
        for (int i = 1; i < marks.length; i++) {
            if (marks[i] < smallest) smallest = marks[i];
        }
        System.out.println("Smallest: " + smallest);
    }
}
```

7. Input: Fixed input values are declared in the program: marks = {45, 67, 80, 52, 90}.

8. Output:
Smallest: 45

Time / Space Complexity: O(n) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- Which comparison finds a smaller value?
- Why is initialization important?
- Can this work with negative values?

Experiment No. 2.3 — Count values greater than 60

1. Experiment Title: Count values greater than 60

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A traversal can track the current minimum/maximum or count elements meeting a condition.

5. Algorithm / Procedure:
Set count to zero; inspect each element; increment count when the element is greater than 60; print count.
Then compile and execute the Java program.

6. Program:

```
public class Task_03_Count_values_greater_than_60{
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int count = 0;
        for (int value : marks) {
            if (value > 60) count++;
        }
        System.out.println("Count: " + count);
    }
}
```

7. Input: Fixed input values are declared in the program: marks = {45, 67, 80, 52, 90}.

8. Output:
Count: 3

Time / Space Complexity: O(n) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is a counter?
- Does 60 itself satisfy value > 60?
- What is the difference between count and sum?